

Trends of carbon emission intensity from electricity energy generation across Italian regions

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1. Research question

The workload was distributed equally among team members. The aim of the project is to **compare greenhouse gas emissions between Italian regions**, obtaining an estimate of the effectiveness of decarbonization policies. In particular, we wanted to develop a very simple but also valid **metric (carbon emission intensity) for the comparison**, which is non-trivial because regions vary greatly in terms of industrial and population density, transport and climate (and therefore agriculture and domestic heating). For this reason, we decided not to focus on total emissions for the metric but to choose one specific category: **emissions due to electricity generation**.

In fact, these emissions are present in each region because power plants are needed everywhere and the amount of electricity generated by them can be used for normalization. Furthermore, their variation is mainly due to the composition of the regional energy mix and therefore represents a **good proxy for assessing how much a region is committed to promoting renewable and clean technologies**. Normalizing total emissions by electricity generation would instead be incorrect, since it would connect unrelated quantities (emissions from agricultural fertilizers or industrial materials have nothing to do with energy plants). Obviously, both the national electrical system and the collection of emissions data are very complex issues: plants take time to be built, certain technologies' performances depend on the environment, emissions should be considered throughout the entire life cycle of a plant, and it is difficult to collect emissions measurements...

For this reason, we have **relied on authoritative Italian and European sources** for our analysis, while remaining aware that they provide only one perspective on a much more complex system and that the results are therefore only an estimate of our research question.

2. Data understanding

The datasets used for the project are:

- 1) Italian electrical energy generation by source
- 2) European greenhouse gases emissions by sector
- 3) Italian greenhouse gases emissions from energy

Dataset (1) is taken from [Terna](#), Italy's transmission system operator (TSO). Terna is the monopolistic company that operates the high voltage grid and links power producers to distribution companies by providing secure transmission infrastructures and real-time system balancing. The dataset contains data on electricity energy generation (measured in GWh) for the following types of sources: "Fotovoltaico", "Geotermoelettrico", "Termoelettrico", "Idrico" and "Eolico". Data are available from 2005 (the first year with Terna as official Italian TSO) to 2022, since last years still need to be verified before the definitive publication. Data are available at province level (further details on its definition in the data preparation section). We will not go into the classification of sources because it is not part of the project and, given the role that Terna plays in the sector, it is reasonable to assume that it is a reliable source for the actual values of electricity generation.

Dataset (2) is obtained from the European project [EDGAR](#), which conducts a European study broken down by sub-national region examining greenhouse gas (GHG) emissions for each sector. These are measured in ktonCO₂eq using Global Warming Potential Values (GWP-100) from IPCC AR5 and include fossil CO₂ only, CH₄, N₂O and F-gases. Data are available for each European region in different sectors with a time span from 1990 to 2022. A brief description of each sector:

- Dom_Avi: Domestic aviation
- Dom_Ship: Domestic shipping
- Indirect: Indirect N₂O emissions
- Agriculture: Agricultural soils, agricultural waste burning, manure management, indirect N₂O emissions from agriculture, enteric fermentation
- Buildings: Energy for buildings
- Energy: Power generation
- Industry: Oil refineries and Transformation industry, chemical processes, Combustion in manufacturing industry, iron and steel production, non energy use of fuels, fuel exploitation, solvents and products use, non-ferrous metals and non-metallic minerals production
- Transport: Road transportation, railways, pipelines, off-road transport

- Waste: Solid waste incineration, landfills, waste water handling

Dataset (3) is taken from [ISPRA](#) (Istituto Superiore per la Protezione e la Ricerca Ambientale), which is a national public body that performs technical and scientific functions for the Italian State in the areas of environmental monitoring and reporting.

The territorial breakdown of atmospheric emissions estimates at provincial level is carried out by ISPRA every 5 years (becoming 2 years in recent years). In particular we used a database containing GHG emissions only from the energy sector to enrich EDGAR's database, since data on Valle d'Aosta energy were not recorded.

3. Data acquisition

Dataset (1) can be fetched from the Terna site using Terna API. We had to register an account in the [Terna Developer](#) portal to access **API tools** and get an API Management ID. Then, we had to create an "application" to invoke the API. We could only use the "PublicAPI-Plan_Eser_Bronze", which has some key rate limits (1 call per second and 300 calls per day) that we faced in the code by adding a sleep function. When the application status became "active", we get the API key and its secret to access the data. We found details on the API catalogue and how to request data from the documentation section of the site. In summary, we had to use our credentials to create an "access token" of type "bearer" via "OAuth2.0" authorization and call an endpoint. The token is valid for 300 seconds and can be used for all public APIs. Finally, we followed the instructions in the "Generation" section of the site and requested "Electrical Energy By Source" data. The results are JSON files: one file per year per call containing data of all provinces and sectors.

Dataset (2) is directly downloaded from "EDGAR total GHG AR5 NUTS2 by country and sector (1990-2022) (v2.0)", where data are found in the second sheet. Metadata can be found in the first sheet. We want to create a database for regional-based Italian analysis. Therefore, we filter the dataset in a Python script to get only the Italian values, removing also 'Dom_Avi', 'Dom_Ship' and 'Indirect' sectors because they are computed on a general national basis. With these adjustments, the dataset does not contain any NaN values.

Dataset (3) is directly downloaded from "GHG ener_2023.xlsx". There is only one Excel sheet, in which we are interested in "REGIONI" and "Anni" columns. They are transformed into a Pandas dataframe via Python script. Being interested only in the "Valle d'Aosta" row, we have filtered the data

and transformed the schema. In this way we have prepared the data in a shape similar to the other datasets. Unfortunately, there are missing data for certain years in the period covered by our scrutiny. The missing values are marked as "NULL".

4. Data storage

We decided to store our datasets into a **relational database** because of its well-defined schema.

In fact, dataset (2-3) are tables (already structured with their schema) and dataset (1) is as a JSON file (in reality it is a union of JSON files of different years, details in the data preparation section) that is not structured as a hierarchical document, even if its fields could have been easily organised that way, but follows this schema:

```
{
  "year": 2022,
  "production_type": "Lorda",
  "region": "Abruzzo",
  "province": "Chieti",
  "source": "Fotovoltaico",
  "production_GWh": 982.474584
}
```

We decided to use [Neon](#), which is a cloud DBaaS service based on PostgreSQL that offers a free environment (with 0.5GB storage limit) and provides a serverless database without local configuration. We used both the tools of the site and our python notebook (in google colab) to import and edit tables. We used the library sqlalchemy to create an "engine" that allows us to connect to the database and perform the tasks of interest via coding.

We used the "schema" tool of the service to divide databases for the **small data warehouse** that we aimed to create: back tier for raw data and front tier for data after preparation steps. Instead, not all the intermediate steps are stored because we can keep track of them keeping the queries in our code and in the historical SQL Editor section of the site.

According to the **star schema**, we decided to place in the front tier, the fact table containing the processed data and the following dimension tables: years (attributes: years from 2005 to 2022) and regions (attributes: the 20 Italian regions). The fact table can be represented as a bidimensional surface formed by $18 \times 20 = 360$ discrete points representing facts (measurements).

In our case, **a fact is the measure of the Carbon Emission Intensity related to electricity energy generation (measured in ktco2eq/gwh)**. After a proper integration process, it can be derived as the ratio between emissions (ktco2eq) and generation (GWh) and represents the average carbon footprint of the electricity mix (further details in the data analysis section).

5. Data preparation

5.1 Data integration (and data quality)

The initial model is a **heterogeneous model** since data come from different sources and are collected for various and different purposes: analysis on electricity energy demand (Terna) and monitoring of climate targets (EDGAR and ISPRA). To determine the carbon emission intensity from electricity energy generation, we have to integrate the two datasets from **schema level integration** to the phase of **correspondence investigation**, i.e. searching for schema heterogeneities and conflicts and solving them including **data quality assessments** too.

The structure of the dataset (1) can be **remapped from JSON to table format** (suitable for our data storage) from the schema provided by the Terna documentation. We also performed a **composed binary pre-integration** of JSON files from different years into a single JSON file covering all years. This integration is easier than the other ones because we are dealing with homogeneous data. Then, we faced another structural difference across schemas: Terna provides temporal data in long format (year, ..., measure) while EDGAR/ISPRA use wide format (... , measure_y1, measure_y2, ...). The latter does not allow natural join on the time dimension so we decided to perform **pivoting** to align the format and we defined the schema: (year, region, measure).

Time dimension

Terna and EDGAR / ISPRA “year” are different: the former aggregate real-time (hourly) measures while the other two use standard IPCC approaches for the emissions estimate (Activity Data × Emission Factor) from annual/monthly statistical data. Since it is not possible to directly measure most of the emissions daily, it means that the time granularity of the problem is different. However, we can ignore that fact saying that **“year” represents the same temporal object in all datasets** since we are not interested in daily and seasonal variations. This is valid as long as we assume that the year is covered evenly and densely and not that, for example, there are models that use more data of a specific month or day in the week (with consequent seasonal bias). For the integration, we selected all different years of Terna (18 years), all years of ISPRA after 2005 (6 years), leaving NULL values for missing data, and all years of EDGAR after 2005 (18 years).

Space dimension

Terna has to identify the location of each individual electrical plant in a rigorous and standardized way because location is a critical factor for dispatching,

grid security and territorial statistics. For each plant, the producer must provide full address, municipality, province, coordinates (latitude/longitude) and POD code and these data have to be registered on GAUDÌ (Gestione Anagrafica Unica degli Impianti). A plant can supply consumers throughout Italy, but its production is attributed only to the region in which it is physically located. So, unless irregular installations and misidentifications, Terna provides the highest **completeness** for the space dimension.

Instead, the spatial granularity of EDGAR is a “cell” of a global grid of $0.1^\circ \times 0.1^\circ$ (~100 km²) containing data on coordinates (latitude, longitude), IPCC sector and emissions. This means that positions do not derive from local administrative data or directly from plants but are geometrical intersections, made with GIS (*Sistema Informativo Geografico*), between the grid cells and polygons of NUTS2 regions (*Nomenclature of Territorial Units for Statistics*), which is the official classification of the European Union for statistical territorial units (where level 2 is the regional level). Instead, ISPRA does not use geographical cells but administrative classification of NUTS2 (the same as EDGAR).

This means that the spatial granularity of ISPRA is much larger than EDGAR one, which is much larger than Terna one (typical size of a plant). However, we can ignore this discrepancy at regional level and therefore we can say that **“region” represents the same spatial object in all datasets** as long as we assume that EDGAR models for calculating cell emissions are accurate enough to distinguish power plants emission from other types that are present in the same cell. We also agreed to use ISPRA data with the same assumption, especially since it is only used to enrich the missing data from Valle d'Aosta.

After the **semantic accuracy** and completeness of space dimension, we need to talk about **syntactic accuracy and consistency**. Two objects (Provincia autonoma di Bolzano e di Trento) are treated as two regions in EDGAR and two provinces in Terna. This happened because they belong to Trentino-Alto Adige region, which is a “region with special status”. Treating them as regions will not be coherent with the classification of regions that we wanted in our final database so we aggregated them into a single region. Then, we faced the issue of **synonyms**: we manually checked the definition of all 20 regions (then repeated in the dataset) and found out that Valle d'Aosta is named differently. So, we decided to rename EDGAR nomenclature with Terna's one because the latter is an Italian operator and it is more reliable because it uses official ISTAT names.

Other dimensions

Original datasets contain other dimensions that we used to obtain the aggregated tables. In particular, we have to deal with energy source types (Terna) and emission sectors (EDGAR/ISPRA).

We are not interested in the understanding (and the classification) of energy source types because we aggregate all of them to obtain a unique value of generation for province and regional level that is consistent because they are assumed to cover the entire effective generation (given what has been said in the space dimension section).

Instead, for emission sectors, we don't perform an aggregation of all available ones and we have to describe the sector we used. EDGAR and ISPRA use sectors derived from the Intergovernmental Panel on Climate Change (IPCC) emission sectors, which are international standards to make data comparable between countries: Energy (power plants, refineries, heat production, transport, ...), Industry (concrete, ...), Agriculture (fertilizers, ...), Waste and Land use. They can be reprocessed and subdivided differently according to the source.

In our case, metadata inside the file stated that **EDGAR'S Energy sector means only emissions from power generation**, which is exactly what we wanted. We give the same interpretation also to ISPRA data, even if it is not specified that emissions come only from power generation but from a more generic "energy sector", because we accepted to have an information on the Valle d'Aosta that may be estimated not precisely rather than excluding it from comparisons with other regions (for this reason is it possible that its value will be overestimated).

5.2 Data cleaning and normalization

We used "ydata-profiling" library to create and show **automatically generated reports about the main datasets characteristics** (Fig. 1) and "sqlalchemy", "missingno" and "pandas" python libraries for the cleaning queries (here we show the overview and discuss some aspects, further details in the code).

Dataset (1) is initially composed of **13704 records** over 6 variables (year, production_type, region, province, source, generation_gwh). There are no missing values because regions that do not have a certain energy source in a certain year are not even present in the dataset and regions that have a certain source but do not use it registered a value of 0 in generated_gwh for that year. As already said, the lack of some sources is not a problem because we are interested in the emissions generated (and so if a region does not have a source can't also have emissions related to it). During cleaning, we erased production_type = "Lorda" because we

wanted to use actual generation ("Netta") and we grouped source and province data to obtain the aggregated sum of generation_gwh for a certain (year, region). After that, we obtained **360 values representing 18 years and 20 regions**.

Dataset (2) is a **1539 rows x 39 columns** covering NUTS2 mapping of European sub-regions. For each entry, we have the columns: "Substance", "ISO", "Country", "NUTS2", "NUTS2 desc", "Sector", "Y_xxxx", where "xxxx" are all the years from 1990 to 2022. We want to perform an Italy-based analysis, therefore we filter with respect to this country which is called "Italy, San Marino and the Holy See" in the dataset. To avoid redundancy, we drop "ISO", "Country" and "NUTS2" columns. We then check for missing data with the "missingno" library. We note NaN values in "NUTS 2 desc" for "Dom_Avi", "Dom_Ship", "Indirect", "Industry" sectors. That is natural, indeed these are nation-wide measures, therefore they do not refer to any particular region. We want to perform a region-based analysis, therefore we drop these four rows. It is interesting to notice that nation-based "Industry" has a streak of missing data for years from 1990 to 2012 and from 2016 to 2019. The first year span hole is due to the fact that this kind of measure was only at regional-level prior to 2012. However, as we said, this row is dropped. At this stage, we obtained a dataset with 125 rows and 37 columns. The records that do not refer to the energy sector were dropped. We then check for sectors of each region and we find that "Valle d'Aosta/Vallée d'Aoste" is missing "Energy". This fact leads to the need of dataset (3) enrichment, after which we obtained **360 values representing 18 years and 20 regions**. We note that this time we kept some NULL values for missing values of Valle d'Aosta because we decided to avoid replacing values with averages, values given by a regression or copied values since it is non-trivial to assume the real distribution of emissions and this could lead to biased estimation of the indicator for this region.

Dataset (3) is extracted from the downloaded Excel file as a **40 rows x 10 columns dataset**. The first column "REGIONI" records the entry's region while the remaining columns cover years from 1990 to 2021 (with several gaps), under the header "Anni". Under "REGIONI" columns there are also other different aggregations which we are not interested in. Emissions data are recorded as tons of CO2eq. We selected only "Valle d'Aosta/Vallée d'Aoste" values and pivoted the table to homogenise the schema. To remain consistent in units of measure, we multiply each value to 1e-3. In this way, we convert to Ktons of CO2eq. Missing values are marked as "NULL". The result is an **18 rows x 3 columns** table that enriches dataset (2).

6. Data analysis

From exploratory data analysis on cleaned datasets we can **derive lots of insights**: we can fix a year, or few years, and try to **compare the amount of generation** between regions (Fig. 2); we can fix a year and create **2d clusters on generation versus emissions** plain to find similar behaviours between regions (Fig. 3).

Nevertheless, generation and emission depends on many external features (population, technology, ...) and may be difficult to obtain insights that are not a mere description of the phenomenon. For example, Lombardia generate more emissions than Molise. However, this could be due to the fact that it has more inhabitants or industries and therefore needs more power, and not because the electricity mix is composed of more polluting sources. Therefore, the idea is to **normalize these two absolute values to obtain a relative value** that allows for more meaningful comparisons across different years and regions: the **carbon emission intensity** index.

$$C_{y,r} = \frac{\text{Emissions from energy generation}_{y,r}}{\text{Energy generation}_{y,r}}$$

It is a much more interesting indicator than solely emissions or total emissions / generation since total emissions depend on the scale of the system and emissions that are also from non-energy sectors will overestimate the real value and give a result that does not have a clear interpretation. This index allows for more complex analyses but still has some weaknesses. For example, if two regions have the same level of carbon emission intensity with the same mix of sources, this could mean that there are differences in the efficiency and technology used. On the other hand, a region with a lot of industry but little generation may appear clean if we only look at emissions and a region with a lot of thermoelectric may appear dirty even if it has a very efficient economy with low emissions in other sectors.

In our analysis we wanted to answer the following questions: How much has regional carbon emission intensity from power generation changed before, during, and after the advent of decarbonization incentives and new technologies development? What was the speed of this transition and which region has accumulated the most improvements over time? To answer these questions, we used a **rank histogram (Fig. 4) as an indicator of the final result and a choropleth map as an indicator of the “path” taken (Fig. 5).**

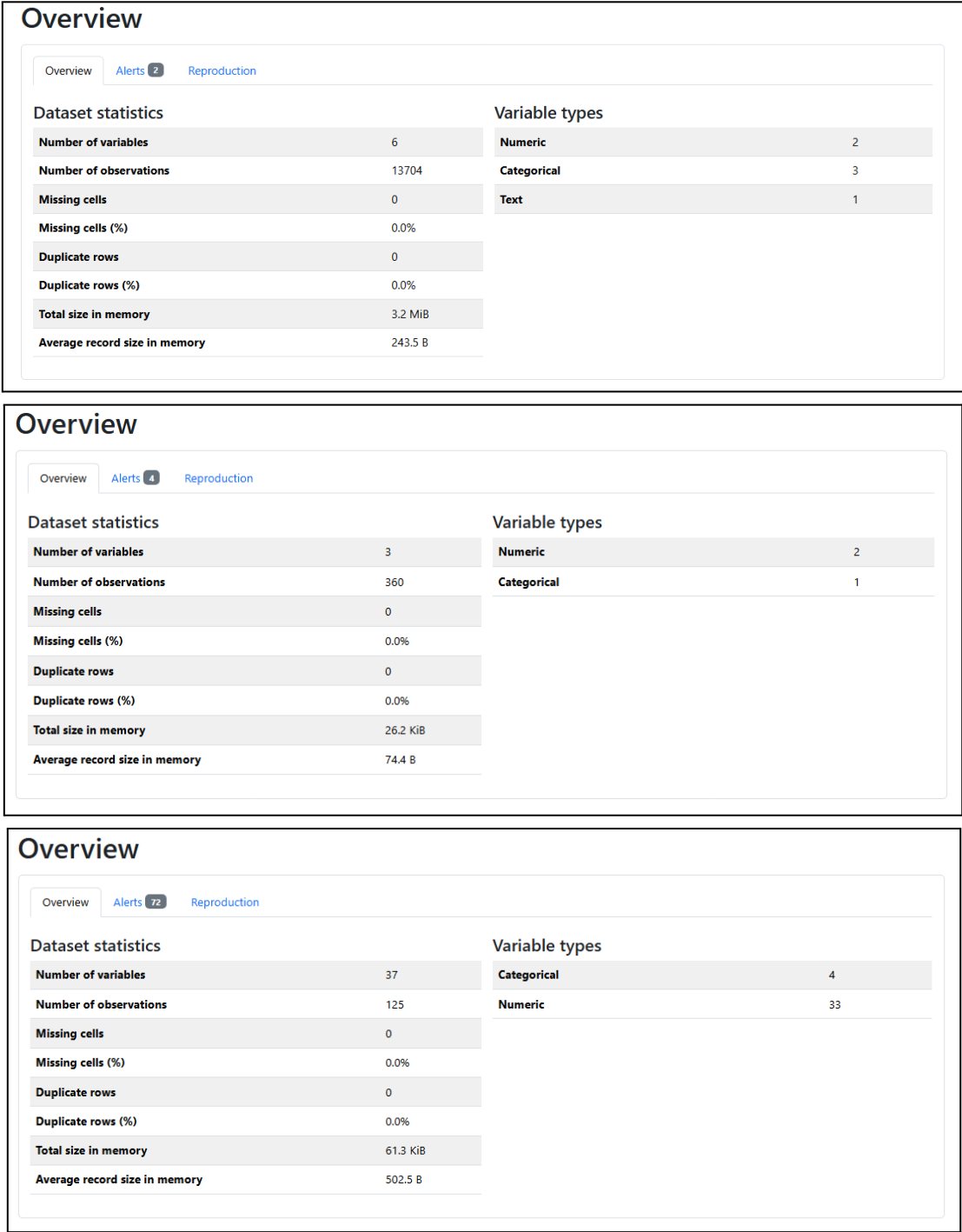
2010 can be considered one of the first years in which, thanks to incentives such as the “Conto Energia”, renewable technologies began to develop.

The “peak” of the emissions problem was reached in 2015, the year in which it was decided to impose a decarbonization target at the Paris conference. Therefore, also taking into account the time needed to build plants and implement strategic plans, we decided to take 2005, 2015, and 2022 as samples to compare the periods. The first graph shows three histograms for the aforementioned years, assessing the absolute value of the index level. Regions are ranked from the least emitting (best) to the most emitting (worst), and their position was marked with numbers beside the histogram bars. This provides an idea of both the absolute improvement of a region over time and the overall improvement of the regional index (value). The second graph shows two maps of Italy coloured with the average and the cumulative distribution of the annual percentage shifts of the index, respectively. This allows us to capture not only the final outcome but also the improvement that each region has achieved during its decarbonization process. This measure is not distorted by a single anomalous year and reflects the consistency of the trend and shows a clear regional difference in different areas of Italy (central Italy regions stand out negatively from the others).

7. Conclusion

We wanted to study greenhouse gas emissions of Italian regions and their decarbonisation policies. We defined the Carbon Emission Intensity, a metric that smooths differences between regions such as population and territorial size, and we focused our analysis on emissions related to electric energy production. Datasets are collected from reliable sources such as Terna, EDGAR and ISPRA. The first two gave us information about electricity generation and greenhouse gas emissions while the latter enriched the EDGAR dataset. We obtained Terna datasets via API while we performed a download for both EDGAR and ISPRA. We chose to use a relational model and transformed our datasets' schemas accordingly. We decided to store the data in a two-tier data warehouse on Neon. Back tier hosts our working datasets while front tier contains our final dataset, tailored to a star schema, and the tables for years' and regions' indexes. Finally, we performed data analysis on both the parent datasets and on the final one. We obtained a simplified yet solid analysis of the decarbonisation policies of each Italian region by using a rank histogram as an indicator of the final result and a choropleth map as an indicator of the “path” taken. In particular, we observe Central Italy to have an increasing trend while the rest of the country, except for Trentino-Alto Adige, Friuli-Venezia Giulia and Liguria, better its environmental impact.

8. Figures



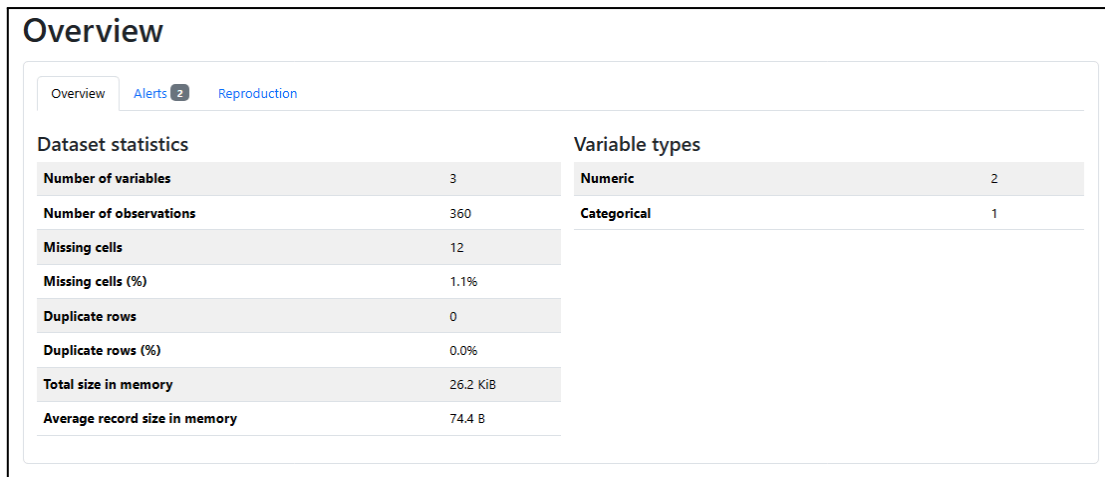


Figure 1: Report overviews generated with ydata-profiling library for Terna and EDGAR datasets. From top to bottom: Terna before cleaning, Terna after cleaning, EDGAR before cleaning, EDGAR after cleaning.

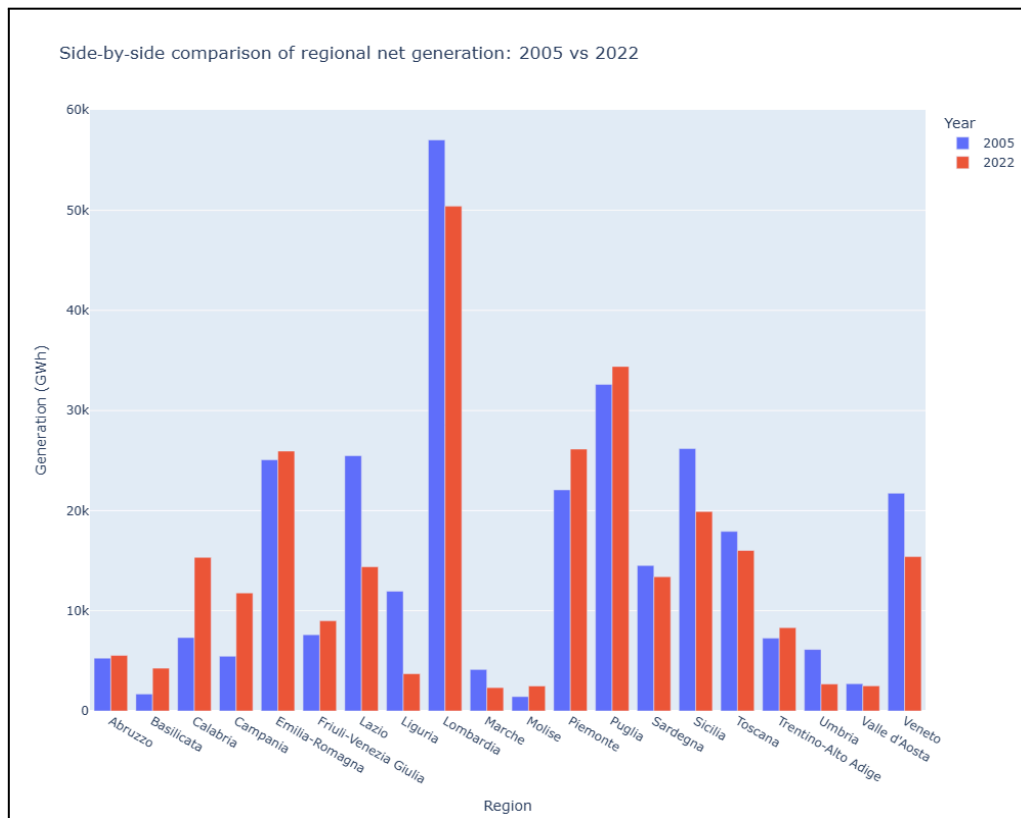


Figure 2: We compared the generated power for different regions, slicing the picture at years 2005 and 2022. We note that the “record” of Lombardy remains consistent in these two years, though its production slightly decreased. Looking just at these data we cannot assert any common trend. Indeed, we have regions with a decrease (such as Lombardia, Veneto, Sicilia and abruptly Lazio), others recorded an increase (as Calabria, Campania and Piemonte) and lastly others with small-to-none variations (as Abruzzo and Valle d’Aosta). However, this plot is useful to grasp the magnitude of the numbers.

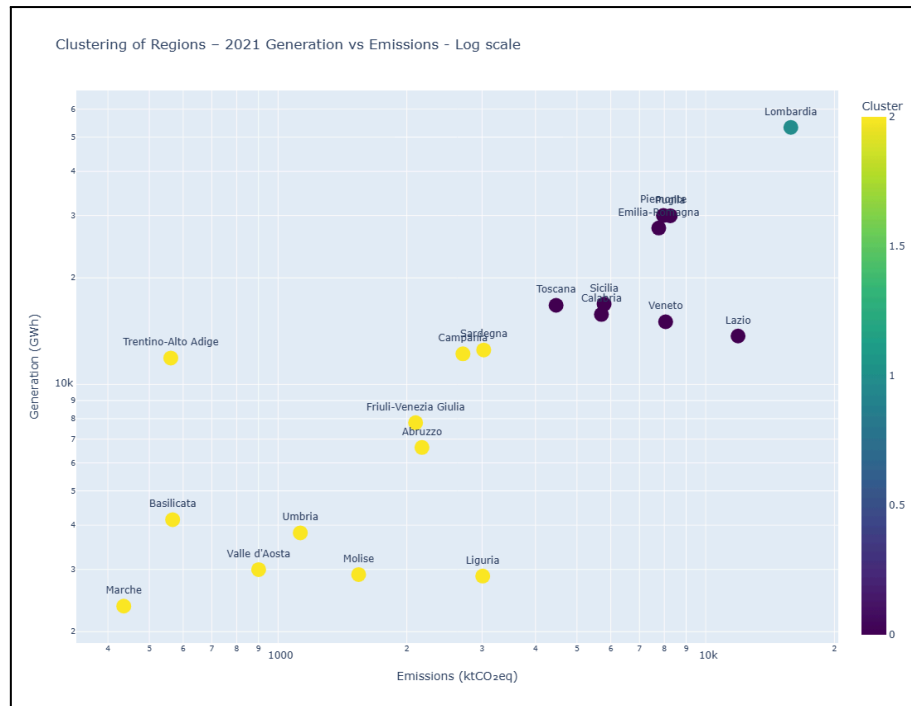


Figure 3: We perform a cluster analysis to look for similarities between regions, considering their energy generation and their emissions (regarding energy sector) as dimensions. We hypothesise that maybe North, Central and South (with Islands) Italy regions may be similar. Therefore, we set the number of clusters to 3. However, we note that it is not clearly the case. Indeed, for example, we observe Sicilia (Island) and Calabria (South) to be in the same cluster of Veneto (North) and Lazio (Central). Lombardia (North) is a clear outlier, having great both generation and emission values. A possible reason for this absence of a trend may be linked to the great disparity in magnitudes between regions.

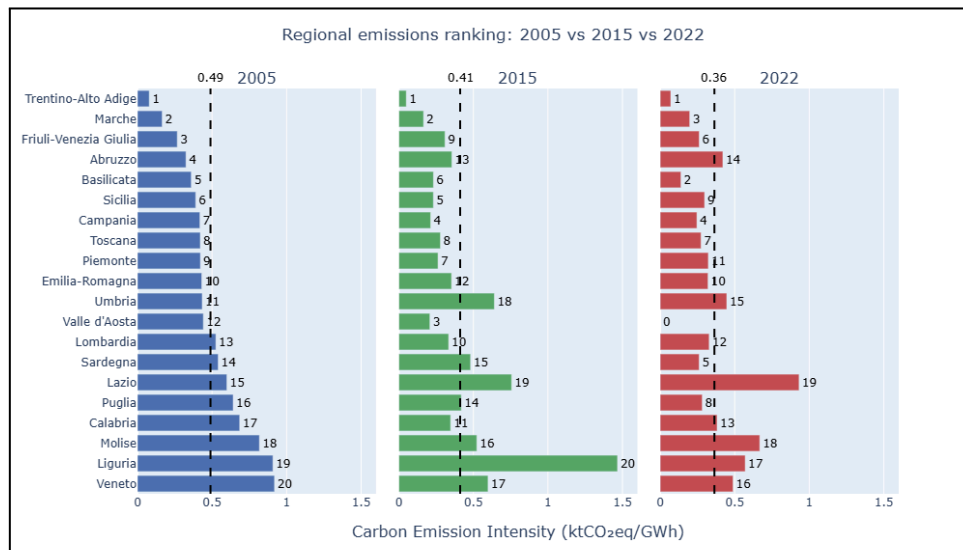


Figure 4: This is a ranking of the twenty Italian regions with respect to the Carbon Emission Intensity. The numbers beside the bars mark the region's position in the ranking of lowest emitting regions. We select three years, in particular, 2005, the beginning of the time span, 2022, the end of the time span, and 2015, the year of Paris' Agreement (starting year of decarbonisation projects). The first snapshot gives us the initial situation, where regions with a lower Carbon Emission Intensity are classified higher. We note how then their positions shift between the years. With respect to the previous clustering, we observe now more normalised quantities. Indeed, Lombardia appears to be in the middle while its raw values in emissions and generation are pretty high compared to the rest of the regions. The "0" for Valle d'Aosta in 2022 is due to missing data. Lastly, the black dashed vertical line marks the mean for each year and it gives us a perspective about the overall trend. We observe a decreasing trend.

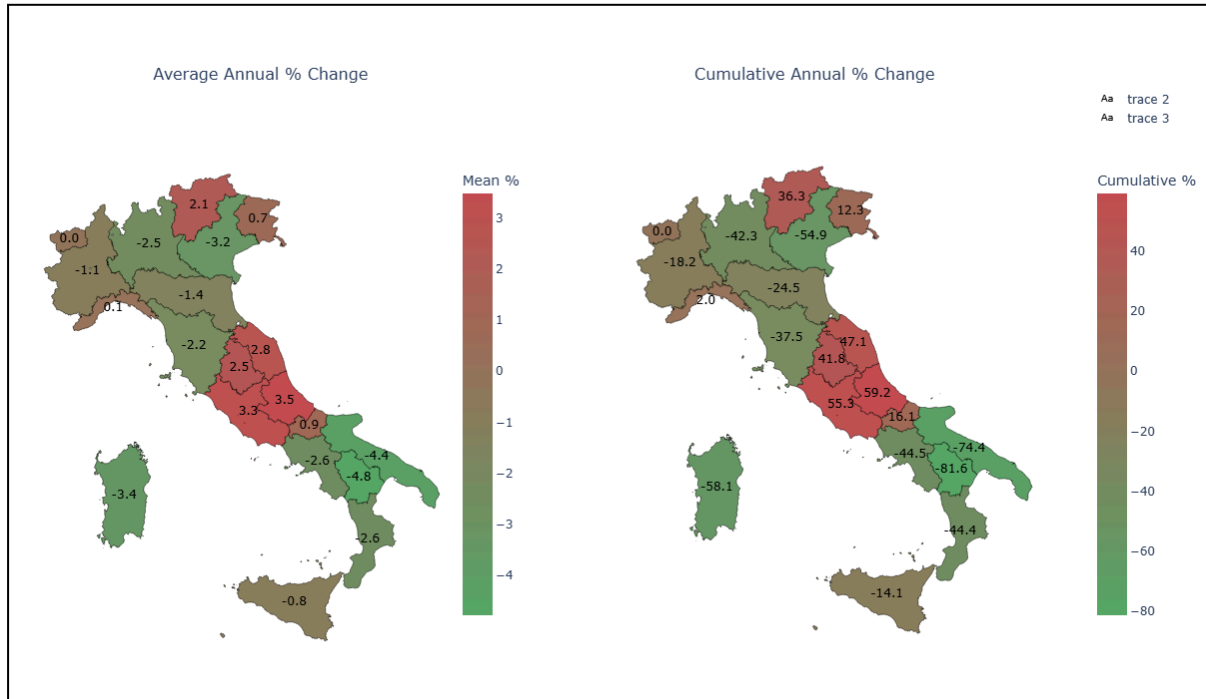


Figure 5: In this final plot, we observe geographically the changes of Carbon Emission Intensity for each region. The first map records shifts between two years averaged for the whole time span, while the second one covers cumulative shifts. We clearly see that Central Italy follows a coherent increasing trend in carbonisation level. The rest of the country, except for Trentino-Alto Adige, Friuli-Venezia Giulia and Liguria, better its environmental impact.